Diseases of the Urinary Tract

NURA 340 Pathophysiology | Lecture Study Guide

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Lecture roadmap: Urinary tract infections: classification and etiology → Cystitis (lower UTI) → Pyelonephritis (upper UTI) → How to tell them apart

After reviewing normal kidney and urinary tract function in the first lecture of this module, this lecture covers what happens when infection develops along the urinary tract. Watch the first lecture before this one. The focus is cystitis (infection of the bladder) and pyelonephritis (infection of the kidneys). Connect the location of the infection and the inflammatory response to the findings you see.

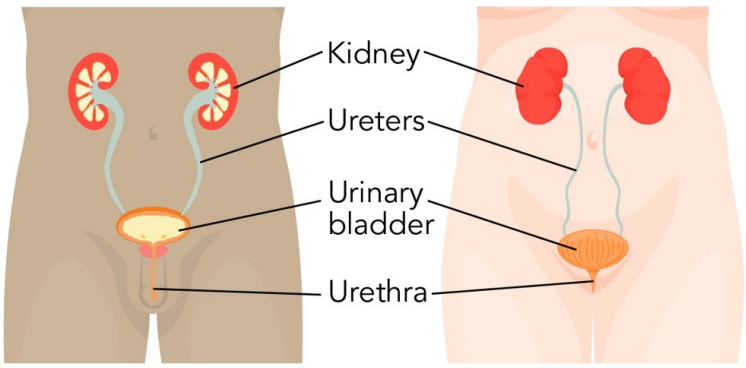
Three questions for each disorder: What allows it to develop? What happens to the affected tissue? How does that explain the symptoms and diagnostic findings? For UTIs, the major distinction is whether the infection stays localized to the bladder or involves the kidneys and produces a more systemic illness.

Learning Objectives

- Etiology, pathogenesis and pathophysiology, clinical manifestations, and diagnostic criteria of urinary tract infections.
- The same for bladder cancer.

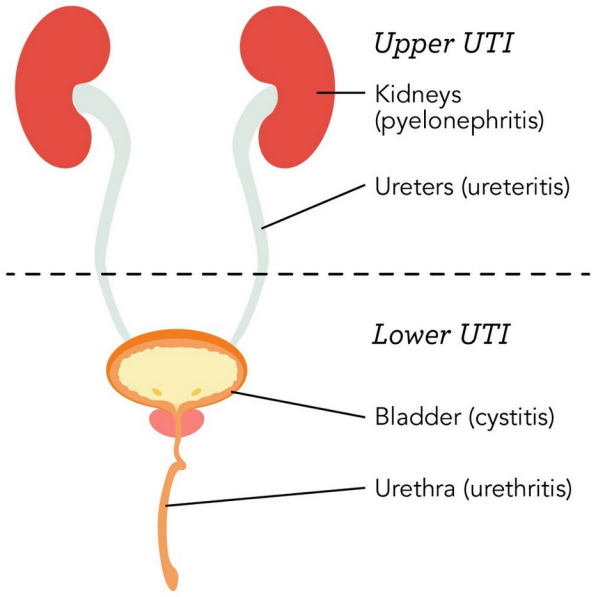
Heads up: Bladder cancer is listed in the slide objectives, but the slides and the lecture do not cover it. Use your ATI Engage module for that topic.

Introduction to Urinary Tract Infections



Urinary tract anatomy: kidneys, ureters, bladder, urethra (lecture slide)

Classification by Location



Upper UTI (kidneys, ureters) vs. lower UTI (bladder, urethra) (lecture slide)

Type	Structures	Infection name
Lower UTI	Bladder, urethra	Cystitis (bladder), urethritis (urethra)
Upper UTI	Kidneys, ureters	Pyelonephritis (kidneys), ureteritis (ureters)

The focus of this lecture is cystitis and pyelonephritis.

Etiology

Most bacterial UTIs begin when bacteria from around the urethral opening enter and move upward. E. coli, which normally lives in the intestinal tract, is the most common organism.

Factor	Why it makes a UTI easier
Shorter urethra	Less distance for bacteria to reach the bladder (from the first lecture)
Catheter	Can introduce bacteria and provides a surface where bacteria attach
Incomplete bladder emptying	Urine left behind gives bacteria an opportunity to multiply
Obstruction (stone, enlarged prostate)	Contributes to urinary stasis, allowing bacteria to multiply. This connects UTIs to other disorders in this module

Discussion Question 1: A patient says they have a "kidney infection" but the diagnosis is cystitis

Think about how to clarify the location of the infection without making the patient feel corrected or dismissed. Most people have heard the terms bladder and kidneys, so explaining the difference between the two organs is a patient-centered way to do it: a bladder infection involves the organ that stores urine (lower tract), while a kidney infection involves the organs that filter blood (upper tract).

Cystitis

Focus: bacterial cystitis, an infection of the bladder (lower UTI).

Etiology

- **Most common:** a bacterial infection that ascends from the urethra into the bladder.
- **Other causes:** chemical irritants, structural anomalies, and radiation. Get a thorough picture of what is going on.
- **Risk factors:** vagina or female anatomy, sexual intercourse (especially with a new partner), certain birth controls such as spermicide (a potential chemical irritant), and personal or family history (first-degree female relative) of UTIs.
- **Males:** cystitis is less common but occurs. Any occurrence is considered more complicated.
- **Very young babies:** a bladder infection is typically more concerning.

Pathogenesis and Pathophysiology

1. **Colonization:** bacteria colonize the urethral area and ascend to the bladder.
2. **Attachment:** bacteria adhere to and invade the bladder mucosa (lining).
3. **Inflammation:** as they multiply they trigger an inflammatory response.
4. **Result:** the bladder lining becomes swollen, erythematous, and painful.

The Engage module has a 3D visualization of cystitis.

Manifestations



Cystitis symptoms: urgent and frequent urination, burning, suprapubic pain (lecture slide)

Manifestation	Why
Dysuria (pain or burning with urination)	Irritated, inflamed bladder lining
Urgency and frequency	Inflammation makes the bladder more sensitive, so the person feels the need to go often even with a little urine to pass
Suprapubic discomfort and bladder spasms	Reflects inflammation in the bladder
Hematuria	The inflamed bladder lining can bleed
Cloudy or foul-smelling urine	Infection in the urine

Pattern to recognize: Are the symptoms localized urinary ones (above)? Or are you seeing systemic symptoms such as fever, flank pain, or the person appearing systemically ill? Those should make you consider infection beyond the bladder.

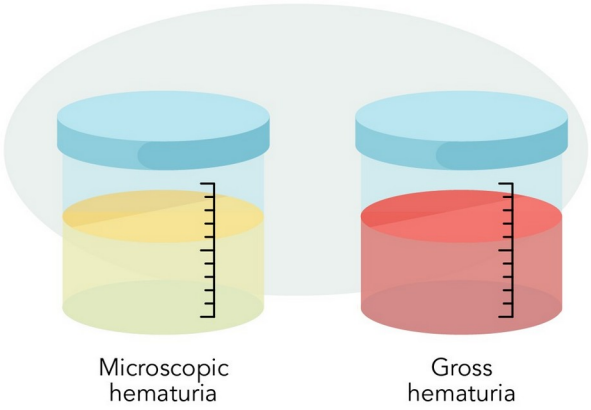
Older Adults

- **Atypical presentation:** new or worsening confusion (especially new onset), agitation, unexplained falls, or general decline in functional status.
- **Caution:** confusion or falls do not automatically mean a UTI, but a UTI should be considered. Either way, do a thorough assessment.

Bacteria in urine is not automatically a UTI: Even when bacteria are present, look for urinary symptoms or systemic signs of infection before attributing a change to a UTI. The way the sample is collected matters: without a clean catch, bacteria from the external genitalia very likely end up in the sample.

Diagnostic Criteria

Diagnosis connects the person's symptoms with evidence from the urine. Nurses are not the ones making the diagnosis; we collect the information.



Microscopic vs. gross hematuria (lecture slide)

Finding	Meaning
Pyuria (white blood cells in urine)	Supports inflammation in the bladder
Leukocyte esterase	A marker associated with white blood cells; may or may not be present
Nitrites	Suggest certain bacteria are present because some organisms convert urinary nitrates to nitrites. Not all bacteria do, so a negative nitrite result does not rule out infection
Hematuria	May reflect irritation of the bladder lining
Urine culture	Identifies the organism and which antibiotic it is susceptible to. Not necessary for every straightforward, uncomplicated bladder infection. Most useful with persistent or recurrent symptoms, an unusual presentation, or concern for a complicated infection

Interpret these findings alongside the symptoms and look at the whole picture.

Impact on the Health of the Client

- **Quality of life:** urgency, frequency, and pain can interrupt sleep and daily activities.
- **Spread:** infection can travel upward to the kidneys, which is more concerning (pyelonephritis).

Connection to pharmacology: Separate medications that address the cause from medications that address the symptoms. Antibiotics target the bacteria. Urinary analgesics (the instructor mentioned an NSAID as an example) can reduce discomfort but do not eliminate the infection. Feeling better and eliminating the underlying cause are related but not the same.

Pyelonephritis

An upper urinary tract infection involving the kidneys.

Etiology

- **Usual route:** bacteria ascend from the bladder through a ureter and reach the kidneys.
- **Obstruction and stasis** make this more likely. Think of someone with stasis or an obstruction.
- **Vesicoureteral reflux:** contributes by allowing urine to flow backward from the bladder toward the kidneys.

Pathogenesis and Pathophysiology

1. **Inflammation:** once bacteria reach the kidney they cause inflammation of the renal pelvis and the kidney tissues.
2. **Local effects:** the kidney location explains flank pain and tenderness near the kidneys on the lower back.
3. **Systemic response:** explains fever, chills, malaise, and nausea and vomiting, which we would not typically see with a lower UTI.

Manifestations



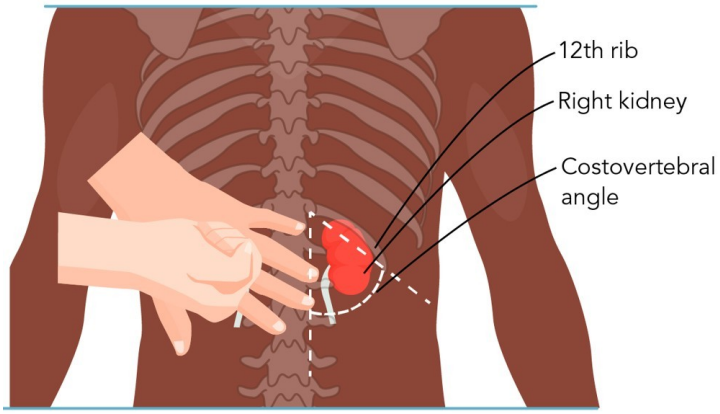
Pyelonephritis symptoms: fever, back or side pain, chills, nausea (lecture slide)

	Cystitis (lower)	Pyelonephritis (upper)
Location of symptoms	Bladder: dysuria, urgency, frequency, suprapubic discomfort	Flank pain and lower back tenderness near the kidneys
Systemic signs	Not typical	Fever, chills, malaise, nausea, vomiting
Urinary symptoms	Present	Dysuria, urgency, or frequency in some people but not always present

Key shift: The change from the localized bladder pattern to findings suggesting kidney involvement. A person does not need to have all of these findings to have pyelonephritis.

Diagnostic Criteria

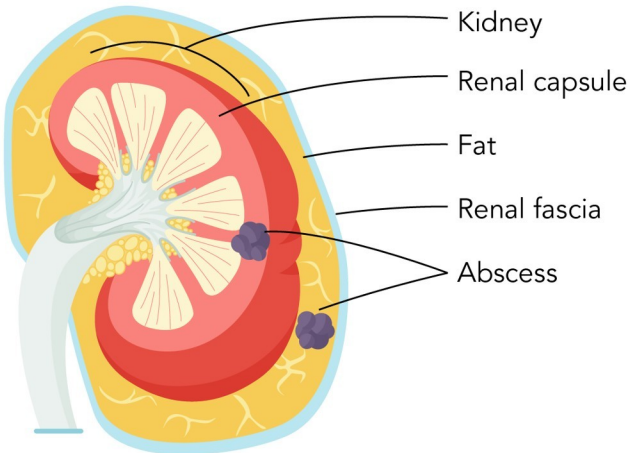
Your role as a nurse is assessment: history, physical findings, and lab results together.



Checking for costovertebral angle (CVA) tenderness (lecture slide)

Assessment	What it shows
CVA tenderness (costovertebral angle, on the back where the kidneys are located)	Supports concern for kidney involvement when it fits the rest of the clinical presentation
Urinalysis	White blood cells and bacteria in the urine
Urine culture	Identifies the organism
Blood tests	Assess the systemic response and kidney function
Imaging	Generally not needed. May be needed if there is concern for obstruction, abscess, or another complication (a provider decides)

Impact on the Health of the Client



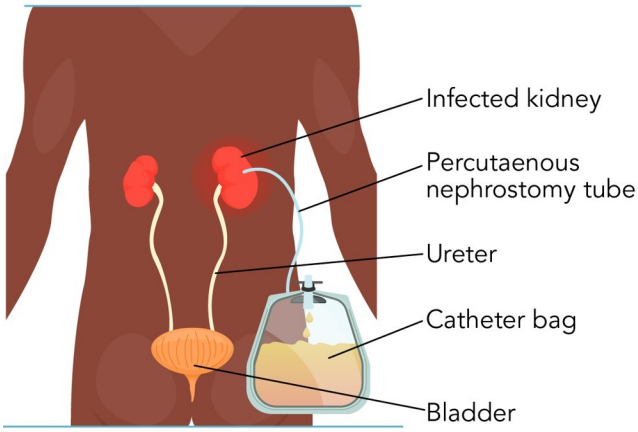
Kidney with abscesses (lecture slide)

Infection in the kidney tissue can have more serious consequences:

- **Acute kidney injury:** relatively common with pyelonephritis.
- **Abscess formation** and sepsis (a huge concern).

- **Renal scarring:** severe or recurrent infection can scar the kidney and decrease kidney function.

Treat adequately and as soon as possible.



Infected kidney with a percutaneous nephrostomy tube draining to a catheter bag (lecture slide)

The slide shows an infected kidney drained by a percutaneous nephrostomy tube. Management was not discussed.

Connection to other concepts: Infection triggers inflammation. Tissue injury can impair kidney function. A systemic response such as sepsis can threaten perfusion to the kidney. It is a vicious cycle.

Discussion Question 2: Which findings suggest a UTI has moved from the lower to the upper tract, and why does it matter?

- **Lower UTI:** burning with urination and frequency (bladder inflammation).
- **New findings:** fever, chills, flank pain, and vomiting raise concern that the infection involves the kidneys. The location of the infection explains the new findings.
- **Why it matters:** kidney involvement carries a greater risk of tissue injury and systemic complications.
- **Hypotension or decreased urine output:** increases concern because it can mean the kidneys are not producing enough urine, and also that they are not receiving enough oxygen-rich blood (both).

Wrapping Up

- **Start with the location:** bladder inflammation explains dysuria, urgency, frequency, and suprapubic discomfort. Kidney involvement adds flank pain, systemic illness, and impaired kidney function.
- **Follow the mechanism:** bacteria reach the tissue, trigger inflammation, and produce findings that reflect the location and severity of the response.

Use that pathway to explain the symptoms rather than memorizing two separate lists.

Quick Review

Condition	Key points
UTI	Most bacterial; E. coli (intestinal flora) most common; ascends from urethral opening. Risks: short urethra, catheter, incomplete emptying, obstruction (stone, enlarged prostate)
Cystitis (lower)	Bacteria adhere to and invade bladder mucosa; inflamed, swollen, painful lining. Dysuria, urgency, frequency, suprapubic pain, bladder spasms, hematuria, cloudy/foul urine. Older adults: new confusion, falls. Pyuria, leukocyte esterase, nitrites (negative does not rule out). Culture for recurrent or complicated cases
Pyelonephritis (upper)	Ascends via ureter; obstruction, stasis, reflux. Flank pain, CVA tenderness, fever, chills, malaise, nausea, vomiting (lower symptoms not always present). UA with WBCs and bacteria, culture, blood tests; imaging if concern for obstruction or abscess. Complications: AKI, abscess, sepsis, renal scarring
Key distinctions	Bacteria in urine is not automatically a UTI (clean catch matters). Antibiotics treat the cause; analgesics treat symptoms. Systemic signs suggest infection beyond the bladder

Notes on Transcript Cleanup

The transcript had no major garbles in this lecture. Short fragments split across lines ("E." / "coli") were rejoined. Discussion Question 1 and 2 summaries combine the instructor's prompts with material she covered; she did not give a worked answer for Question 1. The slide icons for cystitis include fever, but the instructor said fever and flank pain suggest infection beyond the bladder, so fever is listed under pyelonephritis. The lecture speaker is not named on the title slide; the byline follows the previous renal lectures. Bladder cancer appears in the learning objectives but was not covered in the slides or the lecture.